

Benchmark #2

500 keV Fe on Fe

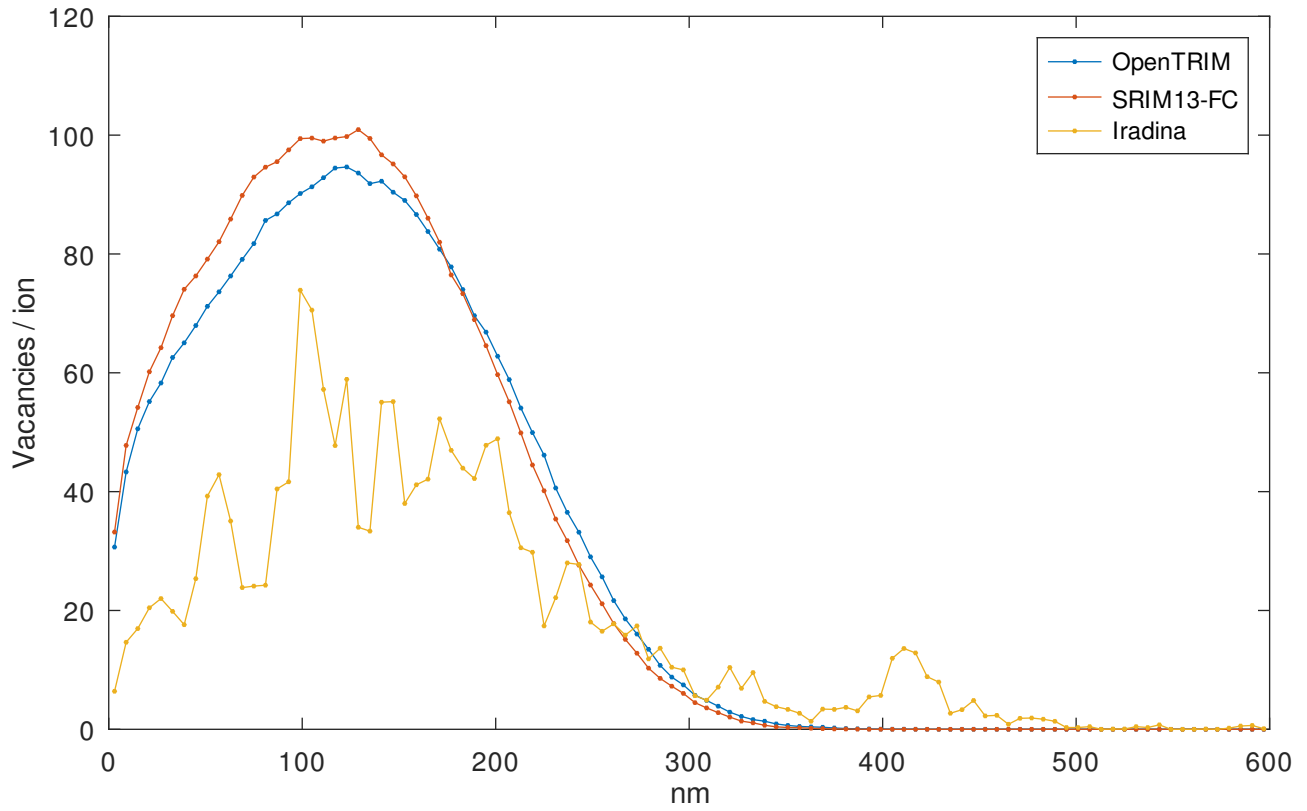
Ion energy $E_0 = 500000$ eV

Target depth = 600 nm

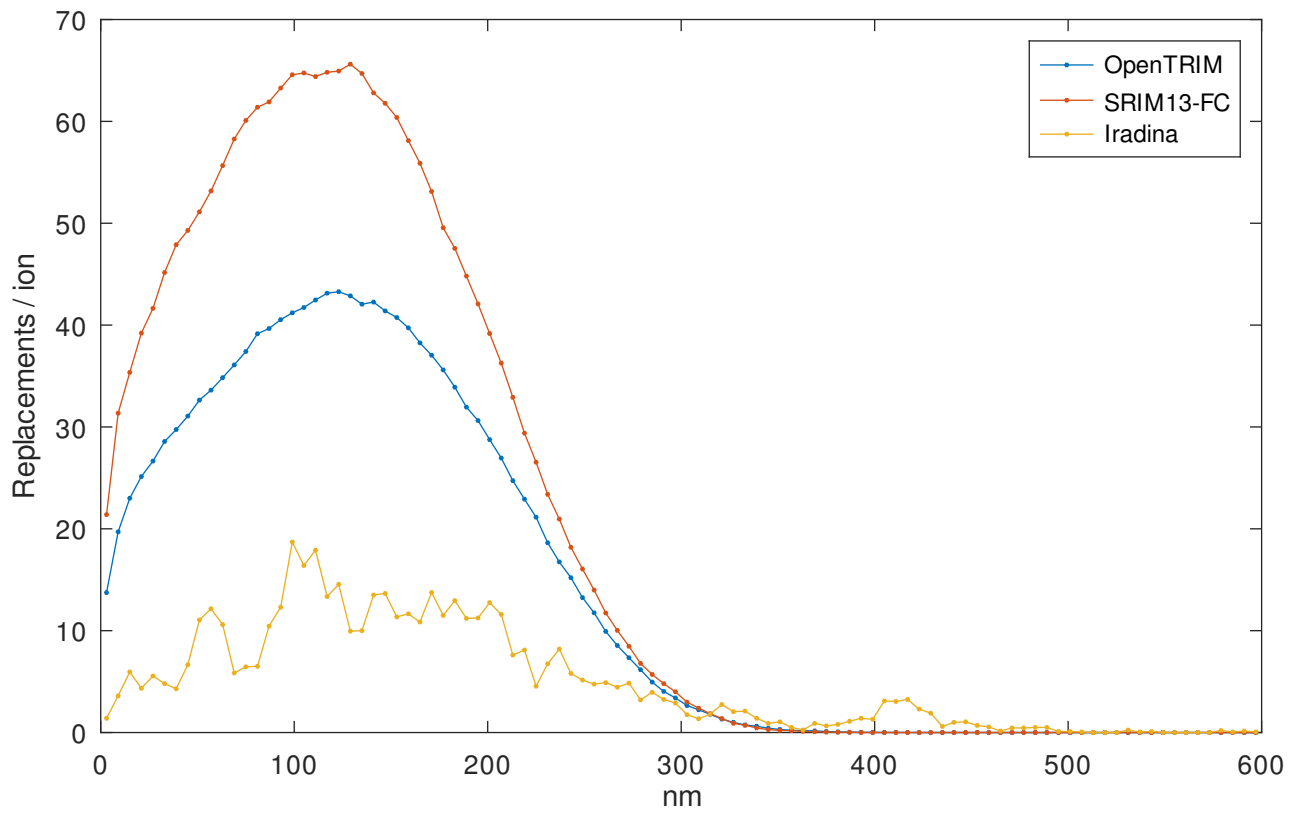
Summary Table

Quantity	OpenTRIM	SRIM13-FC	Iradina
$V(\text{Fe})$	3.1e+03	3.21e+03	1.8e+03
$R(\text{tot})$	1.42e+03	2.09e+03	480
$I(\text{Fe})$	0.999	0.999	1
$\Delta E_e(\text{Fe})/E_0$	0.268	0.262	0
$\Delta E_e(r)/E_0$	0.158	0.155	0
$\Delta E_e(\text{tot})/E_0$	0.426	0.417	0.183
$\Delta E_{ph}(\text{Fe})/E_0$	0.00601	0.00544	0
$\Delta E_{ph}(r)/E_0$	0.567	0.562	0
$\Delta E_{ph}(\text{tot})/E_0$	0.573	0.567	0.215
$1 - (\Delta E_e + \Delta E_{ph})/E_0$	0.00111	0.0157	0.602

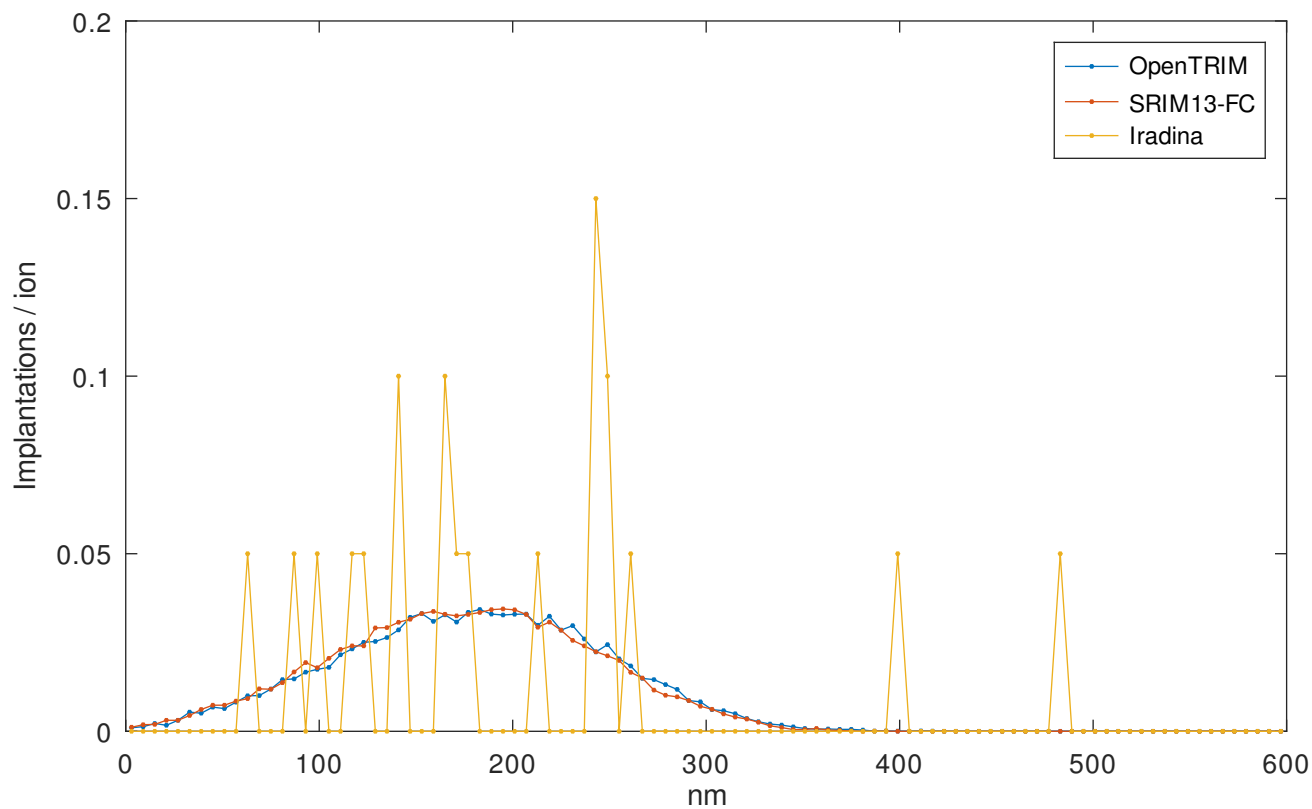
Vacancies of Fe in Fe



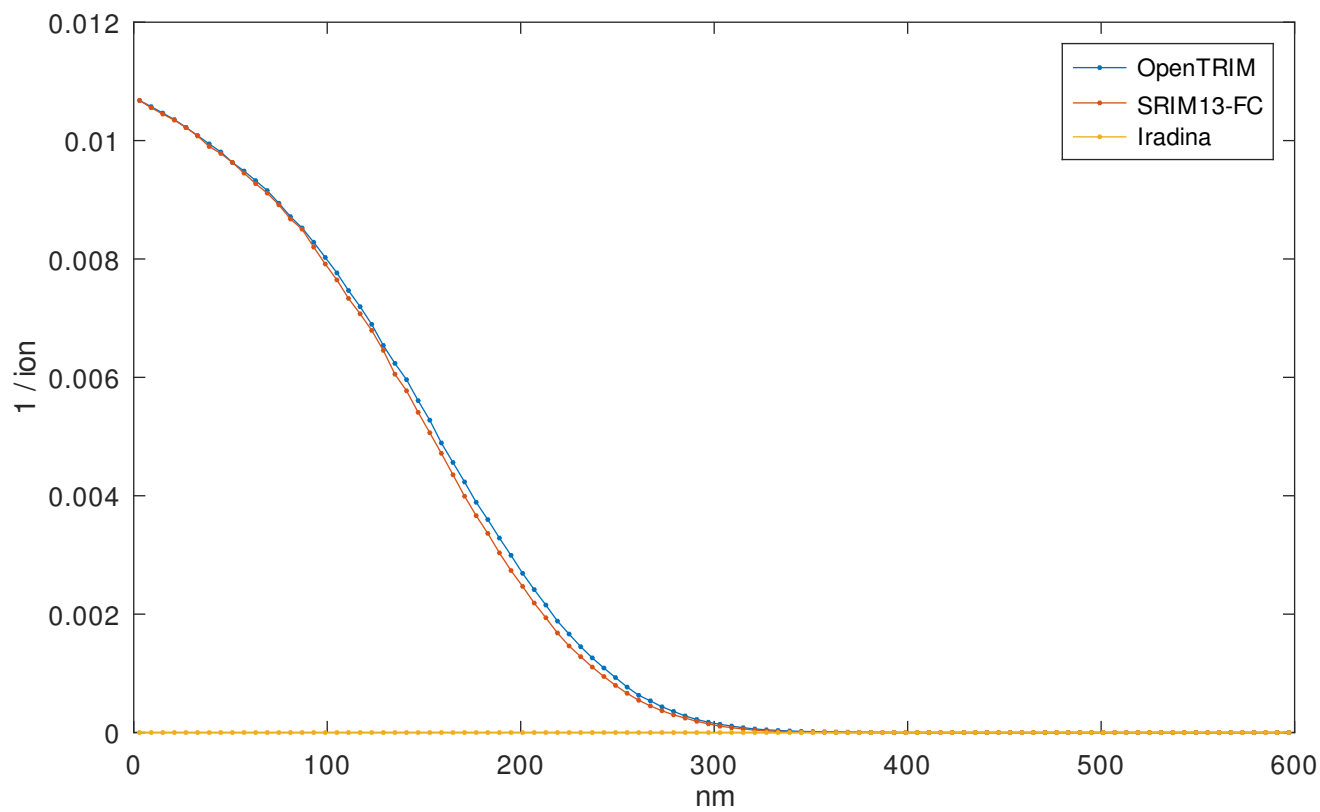
Replacements



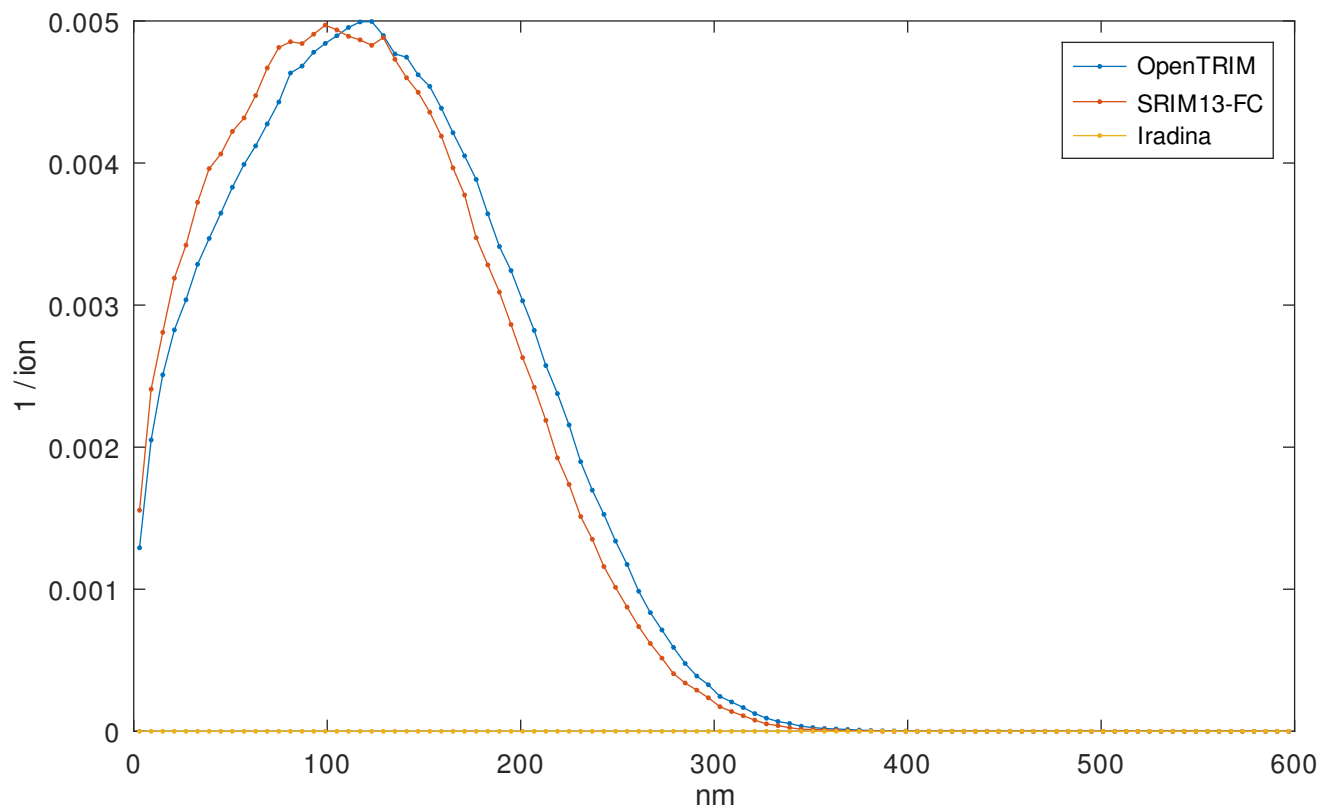
Implanted Fe ion



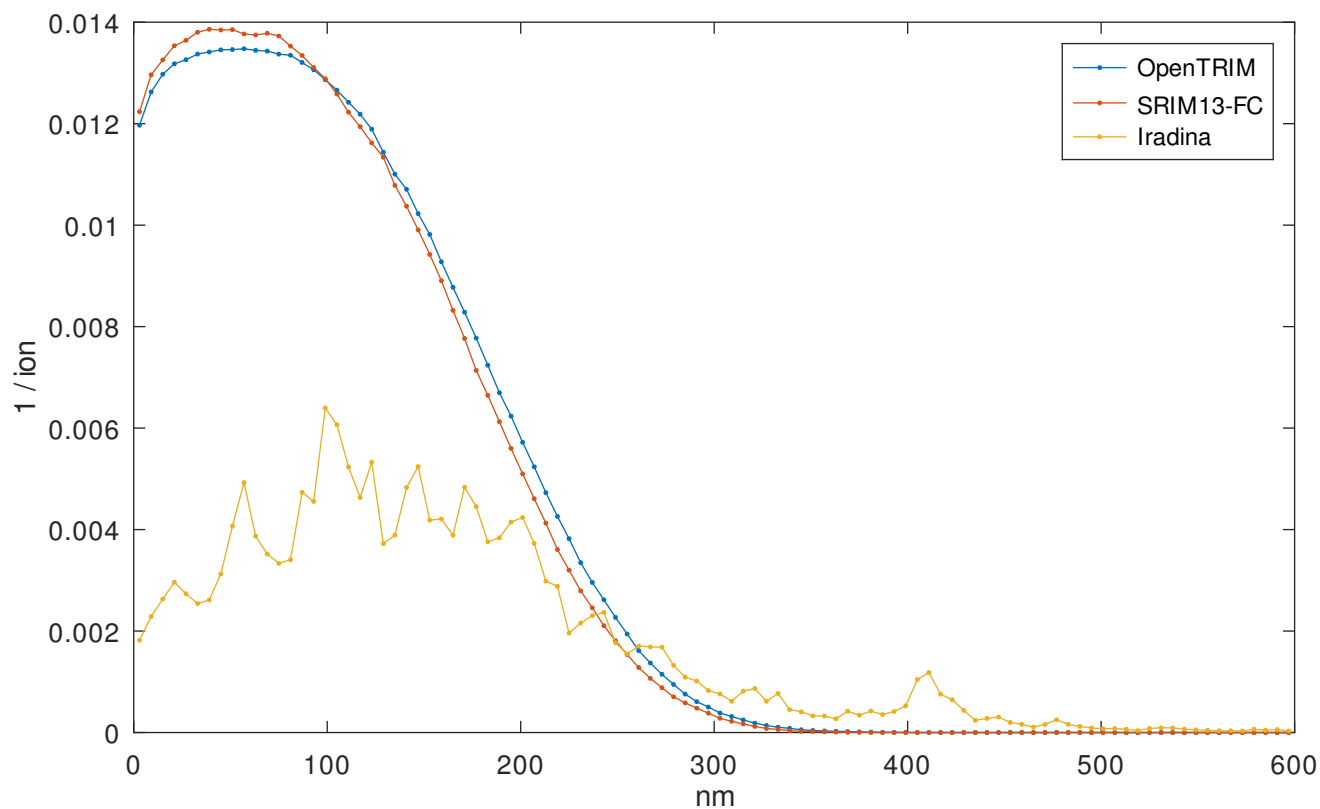
Electronic energy loss fraction $\Delta E_e/E_0$ by Fe ion



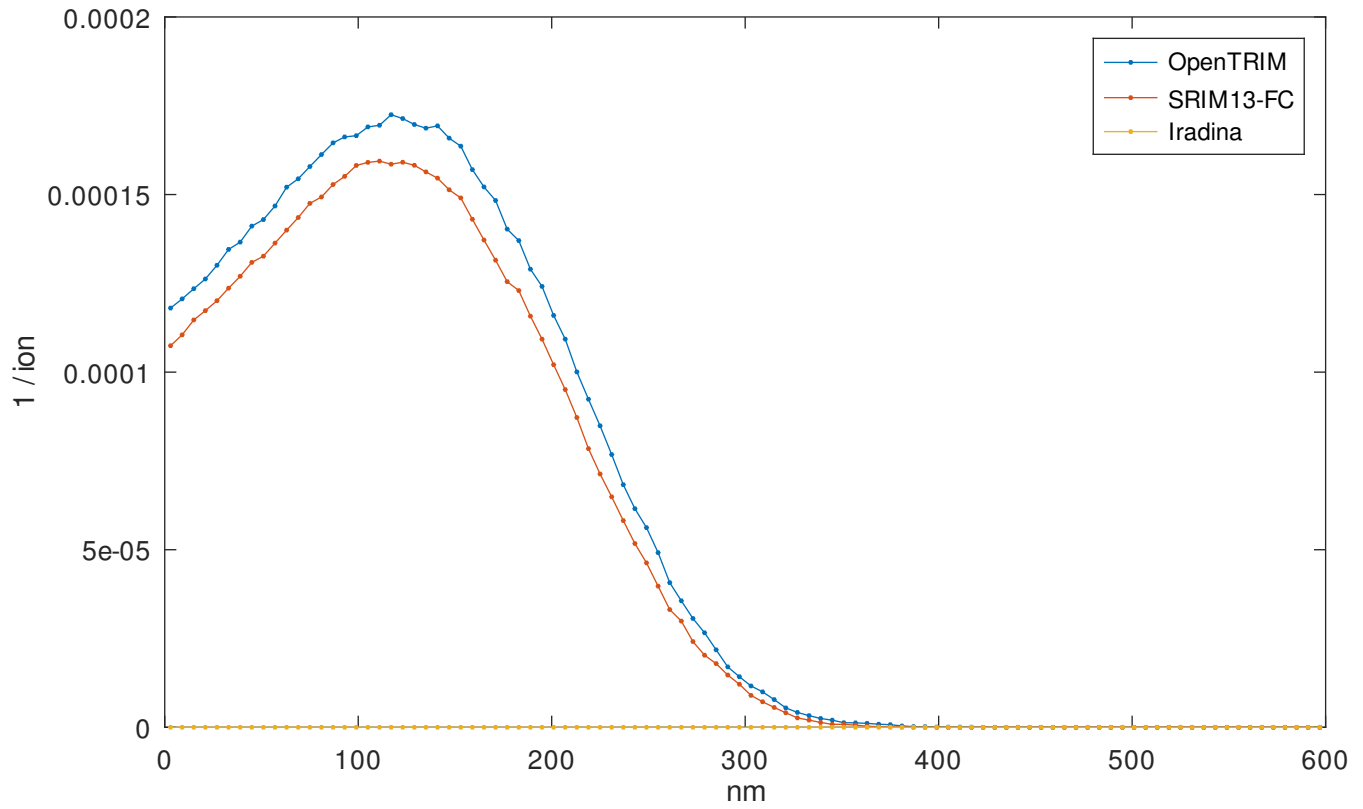
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



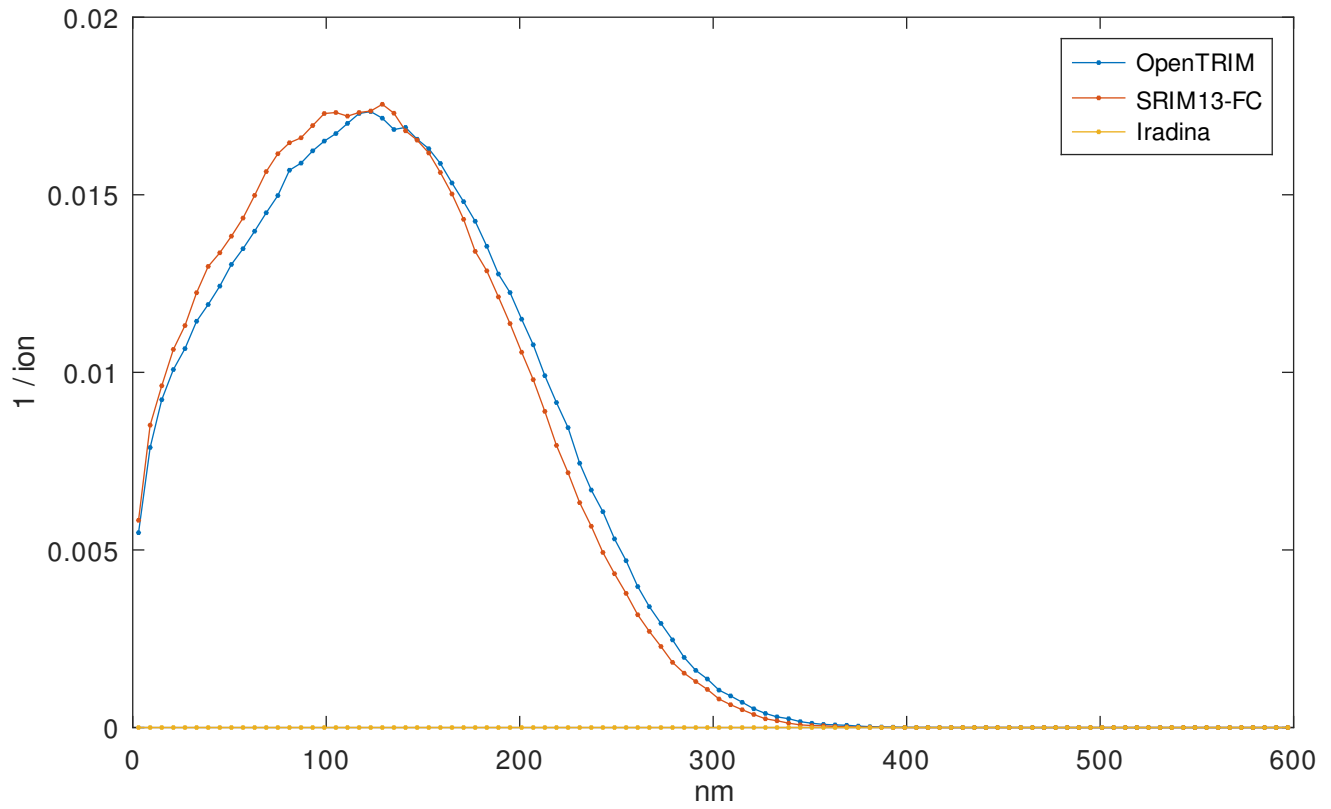
Total Electronic energy loss fraction $\Delta E_e/E_0$



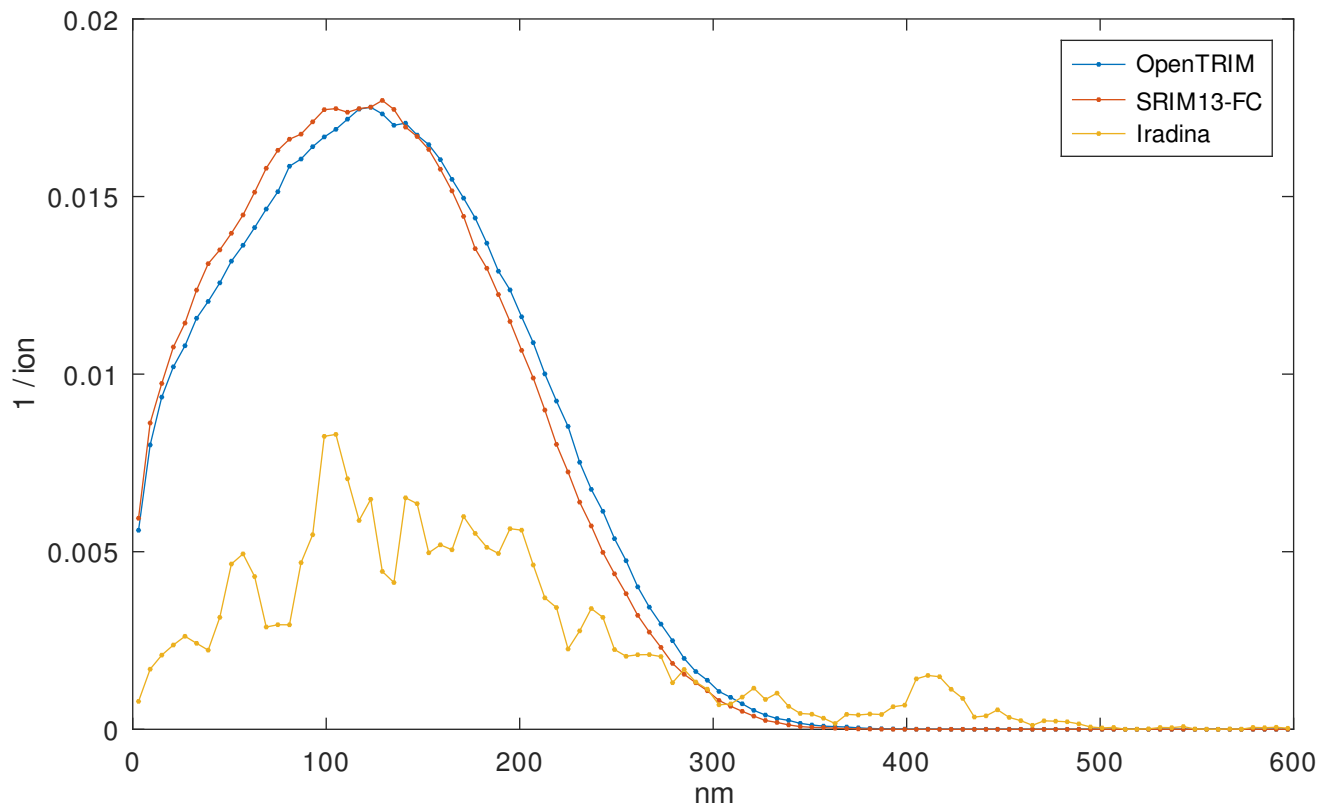
Phonon energy fraction $\Delta E_{Ph}/E_0$ by Fe ion



Phonon energy fraction $\Delta E_{Ph}/E_0$ by recoils



Total Phonon energy fraction $\Delta E_{Ph}/E_0$



Total fractional energy deposition $(\Delta E_e + \Delta E_{Ph})/E_0$

